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Failsafe piston pressure path

Abstract

A piston assembly is disclosed herein. The piston assembly includes a piston having a first end, an opposing second end, and a sidewall extending from the first end to the second end, the piston further including a cavity at least partially defined by the sidewall, a spring guide having a first end and an opposing second end, the spring guide disposed within the cavity of the piston and adjacent the sidewall, an insulator having a first portion extending from the second end of the piston and a second portion disposed within the piston, an insulator shield disposed adjacent the first portion of the insulator, and a component coupled to the second end of the spring guide, the component being offset from the insulator shield by a distance.

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Background/Summary

FIELD

(1) The present disclosure generally relates brake assemblies and more specifically to piston assemblies within brake assemblies.

BACKGROUND

(2) During a hot brake rejected takeoff, the heat sink of a wheel and brake assembly tends to get extremely hot. Brake assembly parts that are in contact with the heat sink of the brake assembly may also get extremely hot and may lose their structural integrity. This may cause components to fail in response to the excessive heat in the brake assembly. For example, the piston assembly, as part of the brake assembly, may fail due to the heat transferred from the heat sink. There are requirements to maintain a static torque on the brake assembly which entails braking pressure to be applied to the heat sink even when the brake assembly is overheating.

SUMMARY

(3) A piston assembly is disclosed herein. The piston assembly includes a piston having a first end, an opposing second end, and a sidewall extending from the first end to the second end, the piston further including a cavity at least partially defined by the sidewall, a spring guide having a first end and an opposing second end, the spring guide disposed within the cavity of the piston and adjacent the sidewall, an insulator having a first portion extending from the second end of the piston and a second portion disposed within the piston, an insulator shield disposed adjacent the first portion of the insulator, and a component coupled to the second end of the spring guide, the component being offset from the insulator shield by a distance.

(4) In various embodiments, the insulator is a first insulator and the piston assembly further includes a second insulator disposed adjacent the second end of the first insulator, wherein the insulator shield includes an opening through which the second insulator extends. In various embodiments, the component is a first component and the piston assembly further includes a second component disposed between the first end of the spring guide and the first end of the piston and coupled to the sidewall of the piston, the second component configured to provide support to the spring guide. In various embodiments, the second component comprises a high temperature capable insulator material.

(5) In various embodiments, the component comprises a high temperature capable insulator material. In various embodiments, the component is press fit onto the second end of the spring guide. In various embodiments, the piston assembly further includes a spring disposed between the first end of the spring guide and the first end of the piston, a lock ring disposed in the sidewall of the piston and configured to prevent the spring guide from extending out of the second end of the piston, and an insulator support disposed within the piston and configured to secure the insulator to the piston.

(6) Also disclosed herein is a brake assembly. The brake assembly includes a plurality of stator disks, a plurality of rotor disks interleaved with the plurality of stator disks, a pressure plate disposed adjacent one of the plurality of rotor disks, and a piston assembly disposed adjacent the pressure plate. The piston assembly includes a piston having a first end, an opposing second end, and a sidewall extending from the first end to the second end, the sidewall at least partially defining a cavity, a spring guide having a first end and an opposing second end, the spring guide disposed within the cavity of the piston and adjacent the sidewall, an insulator having a first portion extending from the second end of the piston and a second portion disposed within the piston, an insulator shield disposed adjacent the first portion of the insulator adjacent the pressure plate, and a component coupled to the second end of the spring guide, the component being spaced from the insulator shield by a distance.

(7) In various embodiments, the insulator is a first insulator and the piston assembly further includes a second insulator disposed adjacent the second end of the first insulator, wherein the insulator shield includes an opening through which the second insulator extends. In various embodiments, the second insulator is configured to engage the pressure plate to apply a braking force. In various embodiments, the component is a first component and the piston assembly further includes a second component disposed between the first end of the spring guide and the first end of the piston and coupled to the sidewall of the piston, the second component configured to provide support to the spring guide.

(8) In various embodiments, the second component comprises a high temperature capable insulator material. In various embodiments, the component comprises a high temperature capable insulator material. In various embodiments, the component is threaded onto the second end of the spring guide. In various embodiments, the component is press fit onto the second end of the spring guide. In various embodiments, the piston assembly further includes a spring disposed between the first end of the spring guide and the first end of the piston, a lock ring disposed in the sidewall of the piston and configured to prevent the spring guide from extending out of the second end of the

piston, and an insulator support disposed within the piston and configured to secure the insulator to the piston.

(9) In various embodiments, the insulator shield is configured to engage the pressure plate to apply a braking force. In various embodiments, the component is configured to engage the pressure plate in response to the insulator shield collapsing.

(10) Also disclosed herein is a piston assembly. The piston assembly includes a hollow cylindrical piston having a first end and an opposing second end, a spring guide having a first end and an opposing second end disposed within the hollow cylindrical piston, a component coupled to the second end of the spring guide, an insulator coupled to the second end of the hollow cylindrical piston, the insulator offset from the component by a first distance, and an insulator shield coupled to the insulator and offset from the component by a second distance.

(11) In various embodiments, the piston assembly further includes a second insulator coupled to the insulator and offset from the component by a third distance and a second component coupled to the piston and configured to provide support for the spring guide.

(12) The foregoing features and elements may be combined in any combination, without exclusivity, unless expressly indicated herein otherwise. These features and elements as well as the operation of the disclosed embodiments will become more apparent in light of the following description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter of the present disclosure is particularly pointed out and distinctly claimed in the concluding portion of the specification. A more complete understanding of the present disclosure, however, may best be obtained by referring to the following detailed description and claims in connection with the following drawings. While the drawings illustrate various embodiments employing the principles described herein, the drawings do not limit the scope of the claims.

(2) FIG. 1A illustrates an exemplary aircraft having a brake system, in accordance with various embodiments.

(3) FIG. 1B illustrates a cross-sectional view of a brake assembly, in accordance with various embodiments.

(4) FIG. 2 illustrates a cross-sectional view of a piston for use in a brake assembly, in accordance with various embodiments.

(5) FIG. 3 illustrates a cross-sectional view of a piston for use in a brake assembly, in accordance with various embodiments.

(6) FIG. 4 illustrates a cross-sectional view of a piston for use in a brake assembly, in accordance with various embodiments.

(7) FIG. 5 illustrates a cross-sectional view of a piston for use in a brake assembly, in accordance with various embodiments.

DETAILED DESCRIPTION

(8) The following detailed description of various embodiments herein makes reference to the accompanying drawings, which show various embodiments by way of illustration. While these various embodiments are described in sufficient detail to enable those skilled in the art to practice the disclosure, it should be understood that other embodiments may be realized and that changes may be made without departing from the scope of the disclosure. Thus, the detailed description herein is presented for purposes of illustration only and not of limitation. Furthermore, any reference to singular includes plural embodiments, and any reference to more than one component or step may include a singular embodiment or step. Also, any reference to attached, fixed,

connected, or the like may include permanent, removable, temporary, partial, full or any other possible attachment option. Additionally, any reference to without contact (or similar phrases) may also include reduced contact or minimal contact. It should also be understood that unless specifically stated otherwise, references to “a,” “an” or “the” may include one or more than one and that reference to an item in the singular may also include the item in the plural. Further, all ranges may include upper and lower values and all ranges and ratio limits disclosed herein may be combined.

(9) The brake assembly, and more specifically, the piston assembly disclosed herein maintains brake pressure longer during a hot brake rejected take off (HBRTO) while retaining structural strength of the piston assembly. For most braking applications, it is beneficial to reduce the number of contacts that allow heat transfer into the piston including into the piston housing and/or hydraulic fluid. Therefore, an additional path for all braking applications is not desirable. In various embodiments, an additional part and/or feature is added to the piston assembly to act as a secondary load path for applying brake pressure to the heat sink in the event that the primary load path fails due to excessive heat. For example, as the piston assembly heats up, the piston assembly may collapse or mushroom in response to being in contact with an overheated heat sink. The piston assembly may transfer brake pressure to the insulator of the piston assembly in response to the collapse or mushroom. In various embodiments, the additional part and/or feature provide a secondary contact point for when the primary piston contact fails to a certain degree and may allow the piston to retain some stroke and keep the brake pressure applied to the heat sink. In various embodiments, the secondary load path may apply brake pressure directly to the heat sink.

(10) In various embodiments, the secondary load path bypasses the main forms of failure in the primary path to ensure that brake pressure is maintained. In various embodiments, the secondary load path does not contact the hotter portion of the primary path until the primary path begins to fail in response to high temperature load. In various embodiments, the secondary load path allows for minimal piston collapse and continued brake application during HBRTO with little to no affect to the heat transfer for other stopping conditions. In various embodiments, this may allow for cheaper, lighter, and/or weaker materials to be used for the piston and/or piston insulator for normal stopping conditions while still meeting HBRTO standards using the secondary load path.

(11) Referring to FIG. 1A, in accordance with various embodiments, an aircraft **10** is illustrated. The aircraft **10** includes landing gear, which may include a left main landing gear **12**, a right main landing gear **14** and a nose landing gear **16**. The landing gear support the aircraft **10** when it is not flying, allowing the aircraft **10** to taxi, take off and land without damage. While the disclosure refers to the three landing gear configurations just referred, the disclosure nevertheless contemplates any number of landing gear configurations.

(12) Referring now to FIG. 1B, there is schematically depicted a brake mechanism (or brake assembly) **100** that may be used by the aircraft **10** of FIG. 1A or any other appropriate aircraft. The brake mechanism **100** is mounted on an axle **102** for use with a wheel **104** disposed on and configured to rotate about the axle **102** via one or more bearing assemblies **103**. The wheel **104** includes a hub **106**, a wheel well **108** concentric about the hub **106** and a web portion **110** interconnecting the hub **106** and the wheel well **108**. A central axis **112** extends through the axle **102** and defines a center of rotation of the wheel **104**. A torque plate barrel **114** (sometimes referred to as a torque tube or barrel or a torque plate or back leg) is aligned concentrically with the hub **106**, and the wheel **104** is rotatable relative to the torque plate barrel **114**.

(13) The brake mechanism **100** includes a piston assembly **116**, a pressure plate **118** disposed adjacent the piston assembly **116**, an end plate **120** positioned a distal location from the piston assembly **116**, and a plurality of rotor disks **122** interleaved with a plurality of stator disks **124** positioned intermediate the pressure plate **118** and the end plate **120**. The pressure plate **118**, the plurality of rotor disks **122**, the plurality of stator disks **124** and the end plate **120** together form a brake heat sink or brake stack **126**. The pressure plate **118**, the end plate **120** and the plurality of

stator disks **124** are mounted to the torque plate barrel **114** and remain rotationally stationary relative to the axle **102**.

(14) The torque plate barrel **114** may include an annular barrel or torque tube **128** and an annular plate or back leg **130**. The back leg **130** is disposed at an end distal from the piston assembly **116** and may be made monolithic with the torque tube **128**, as illustrated in FIG. 1B, or may be made as a separate annular piece and suitably connected to the torque tube **128**. The torque tube **128** has a plurality of circumferentially spaced and axially extending splines **132** disposed on an outer surface of the torque tube **128**. The plurality of stator disks **124** and the pressure plate **118** include notches or stator slots **134** on an inner periphery of the disks and the plate for engagement with the splines **132**, such that each disk and the plate are axially slidable with respect to the torque tube **128**.

(15) The end plate **120** is suitably connected to the back leg **130** of the torque plate barrel **114** and is held non-rotatable, together with the plurality of stator disks **124** and the pressure plate **118**, during a braking action. The plurality of rotor disks **122**, interleaved between the pressure plate **118**, the end plate **120** and the plurality of stator disks **124**, each have a plurality of circumferentially spaced notches or rotor lugs **136** along an outer periphery of each disk for engagement with a plurality of torque bars **138** that is secured to or made monolithic with an inner periphery of the wheel **104**.

(16) An actuating mechanism for the brake mechanism **100** includes a plurality of piston assemblies, including the piston assembly **116**, circumferentially spaced around an annular piston housing **156** (only one piston assembly is illustrated in FIG. 1B). Upon actuation, the plurality of piston assemblies affect a braking action by urging the pressure plate **118** and the plurality of stator disks **124** into frictional engagement with the plurality of rotor disks **122** and against the end plate **120**. Fluid or hydraulic pressure, mechanical springs or electric actuators, among other mechanisms, may be used to actuate the plurality of piston assemblies. Through compression of the plurality of rotor disks **122** and the plurality of stator disks **124** between the pressure plate **118** and the end plate **120**, the resulting frictional contact slows or stops or otherwise prevents rotation of the wheel **104**. The plurality of rotor disks **122** and the plurality of stator disks **124** are fabricated from various materials, such as ceramic matrix composites, that enable the brake disks to withstand and dissipate the heat generated during and following a braking action.

(17) The torque plate barrel **114** is secured to a stationary portion of the landing gear such as the axle **102**, preventing the torque plate barrel **114** and the plurality of stator disks **124** from rotating during braking of the aircraft. The torque tube **128** portion of the torque plate barrel **114** may be attached to the annular piston housing **156** via an annular mounting surface **158**, wherein bolt fasteners **160** secure the torque plate barrel **114** to the annular piston housing **156**. A spacer member or pedestal **162** is positioned between an inner diameter surface **164** of the torque tube **128** and an outer diameter surface **166** of the axle **102**. The pedestal **162** includes a radially inner surface or foot **168** for engaging the axle **102**, a web portion **170** radially outward of the foot **168** and a head portion **172** for engaging the inner diameter surface **164** of the torque tube **128**. The pedestal **162** augments support of the torque plate barrel **114** within the brake mechanism **100** generally and, more particularly, against the axle **102**. The pedestal **162** may be made monolithic with the torque tube **128** portion of the torque plate barrel **114**.

(18) A heat shield **140** is secured directly or indirectly to the wheel **104** between a radially inward surface of the wheel well **108** and the plurality of torque bars **138**. As illustrated in FIG. 1B, the heat shield **140** is concentric with the wheel well **108** and may have a plurality of heat shield sections **142** disposed between respective, adjacent pairs of the plurality of torque bars **138**. The heat shield **140**, or heat shield sections **142**, is spaced from the radially inward surface of the wheel well **108** and secured in place by heat shield tabs **190**, such that the heat shield **140**, or heat shield sections **142**, is disposed generally parallel to the axis of rotation or central axis **112** of the wheel **104** and intermediate the plurality of torque bars **138** and the radially inward surface of the wheel well **108**. In various embodiments, including for heavy-duty applications, the heat shield **140**, or

heat shield sections **142**, may be further secured in place by heat shield carriers **144**.

(19) The plurality of torque bars **138** is attached at axially inboard ends to the wheel **104** by torque bar bolts **146**. The torque bar bolts **146** extend through respective holes in a flange **150** provided on the wheel **104** as shown, which flange **150** for purposes of the present description is intended to be considered as part of the wheel well **108**. Each of the plurality of torque bars **138** may include a pin **152** or similar member at its axially outboard end (i.e., the end opposite the torque bar bolts **146**) that is received within a hole **154** disposed proximate the web portion **110** of the wheel **104**. The heat shield **140**, or heat shield sections **142**, is positioned adjacent a radially inward surface of the wheel well **108** and secured in place by the heat shield tabs **190**.

(20) Referring now to FIG. 2, in accordance with various embodiments, a cross-section of a piston assembly **200** is illustrated. In various embodiments, piston assembly **200** may be an example of piston assembly **116** described above with respect to FIG. 1B. Piston assembly **200** includes a piston **202**, a spring guide **204**, a spring **206**, an air gap **208**, an insulator **210**, and an insulator shield **212**. Piston **202** is cylindrical having a hollow center cavity (e.g., air gap **208**) that is partially defined by side walls of piston **202**. Piston **202** extends from a piston housing and engages a heat sink **214**. As piston **202** engages, and presses into (e.g., the y-direction), heat sink **214** a brake force is applied to the brake assembly (e.g., brake assembly **100**). In various embodiments, heat sink **214** may be an example of pressure plate **118** described above with respect to FIG. 1B. In various embodiments, heat sink **214** may be an outer stator, an end board side of a heat sink, or a stator exposed toward the piston, among others. As heat sink **214** wears down and/or heats up, piston **202** extends further from a piston housing (e.g., in the y-direction). The extension of piston **202** from the piston housing may be called the stroke of the piston. In various embodiments, a warm piston may extend, or have a stroke of, about 5 mm to about 30 mm out the piston housing, and more specifically, about 10 mm to about 20 mm. Generally, as the brakes wear out and warm up during use, piston **202** may extend further from the housing, using more of the available stroke. After a certain stroke (e.g., 20 mm) piston **202** cannot extend further, resulting in reduced braking pressure.

(21) Spring guide **204** and spring **206** are disposed within piston **202**, and more specifically within air gap **208**, and provide additional support for piston **202**. In various embodiments, spring guide **204** and spring **206** provide a buffer, or dampener, for piston **202** if an opposing force is exerted, such as by heat sink **214**, through movement within the brake assembly (e.g., brake assembly **100**). This allows piston **202** to maintain contact with heat sink **214** and provide a consistent brake pressure. Piston assembly **200** further includes a lock ring **216** disposed within piston **202**. In various embodiments, lock ring **216** may be an indent around the circumference of piston **202**. Lock ring **216** may be configured to receive, or seat, an O-ring that secures spring guide **204** in place and prevents spring guide **204** from extending further past lock ring **216** (e.g., in the y-direction).

(22) Insulator **210** is coupled to an end of piston **202**, as illustrated it is the upper end (e.g., in the y-direction). There is an air gap between insulator **210** and spring guide **204** which provides a thermal barrier to reduce the amount of heat transferred from insulator **210** to spring guide **204** during operation. Insulator **210** is secured to piston **202** by an insulator support **218** that is formed within piston **202**. Insulator support **218** prevents insulator **210** from extending past piston **202** (e.g., the y-direction). Insulator **210** has a mushroom shape, with one end of insulator **210** being within piston **202** and the other end of insulator **210** extending past piston **202** and over piston **202**. This provides support for insulator **210** to not be pressed into piston **202** (e.g., the negative y-direction). Insulator **210** is typically made from a high temperature capable insulator material. In various embodiments, insulator **210** may be made from steel, austenitic nickel-chromium-based superalloys such as those sold under the name INCONEL (IN-100 and the like), or another high temperature capable material. That is, a material that is able to maintain structural strength while being subjected to high temperatures.

(23) Insulator shield **212** is coupled to the side of insulator **210** that is facing away from piston **202** (e.g., the y-direction). Insulator shield **210** provides the contact surface for piston assembly **200** to heat sink **214**. Insulator shield **212** is made from a high temperature capable insulator material. In various embodiments, insulator shield **212** may be made from steel, austenitic nickel-chromium-based superalloys such as those sold under the name INCONEL(IN-100), or another high temperature capable material.

(24) Combined, insulator **210** and insulator shield **212** form the primary load path of piston assembly **200**. The force exerted by piston assembly **200** passes through insulator **210** and insulator shield **212**. As a result, the high temperatures from heat sink **214** are passed to insulator **210** through insulator shield **212**. Generally, insulator **210** and insulator shield **212** are capable handling the high temperatures achieved during braking. However, during a hot brake rejected take off (HBRTO) the temperatures within the brake assembly (e.g., brake assembly **100**) and of heat sink **214** may exceed tolerances of piston assembly **200**, specifically, insulator **210** and insulator **212**. This may cause insulator **210** and/or insulator shield **212** to break down, collapse, and/or mushroom in response to the sustained high temperatures. This decreases the effectiveness of the primary load path (e.g., insulator **210** and insulator shield **212**) and may result in brake failure.

(25) With continued reference to FIG. 2, piston assembly **200** further includes a component **220** that acts as a secondary load path. Component **220** is disposed between spring guide **204** and insulator shield **212**. Component **220** is coupled to spring guide **204**, surrounding an end of spring guide **204**. There is an air gap between component **220** and insulator **210** and between component **220** and insulator shield **212**. The air gap provides a thermal barrier between component **220** and both insulator **210** and insulator shield **212** to reduce the amount of heat introduced into piston assembly **200**. Component **220** is separated from insulator shield **212** by a distance $d1$. In various embodiments, distance $d1$ may be about 0.1 mm to about 1 mm. Distance $d1$ provides physical and thermal separation between insulator shield **212** and component **220**. Insulator **210** supports insulator shield **212** and creates distance $d1$.

(26) Component **220** relieves some of the stress experienced by piston assembly **200** in response to piston assembly **200** being exposed to high temperatures, such as during HBRTO. During a hot brake rejected take off (HBRTO) the temperature of the brake assembly (e.g., brake assembly **100**) increases from the friction used during braking. The heat from the brake assembly, including heat sink **214**, is transferred to piston assembly **200**. The increased temperature causes the breakdown of insulator **210** and/or insulator shield **212** and the collapse of piston assembly **200**. Specifically, insulator shield **212** may collapse as insulator **210** collapses. Distance $d1$ is decreased as insulator shield **212** collapses until insulator shield **212** contacts and engages component **220**. Component **220** is cooler than the surrounding insulator **210** and insulator shield **212** because of the air gap of distance $d1$.

(27) In various embodiments, component **220** may be press fit onto spring guide **204**. In various embodiments, component **220** may be threaded and may be threaded onto spring guide **204**. In various embodiments, spring guide **204** may include a bulge, or other protrusion, over which component **220** fits to press fit, or snap, component **220** into place. In various embodiments, a clip may be used that is deformed slightly as component **220** is pressed onto spring guide **204**. The clip then returns to its original shape, securing component **220** to spring guide **204**.

(28) Including component **220** in piston assembly **200** allows piston assembly, and thereby the brake assembly (e.g., brake assembly **100**) to maintain brake pressure longer during a HBRTO. Component **220** allows for minimal collapse of piston assembly **200** while maintaining brake pressure with little to no additional heat transfer during other braking, or stopping, conditions. The minimal collapse of piston assembly **200** and subsequent support by component **220** provides additional braking time for the braking assembly (e.g., brake assembly **100**). Additionally, the use of component **220** as a secondary load path may allow for the use of cheaper, lighter, and/or weaker materials for piston **202**, insulator **210**, and/or insulator shield **212** while still meeting braking

requirements.

(29) Referring now to FIG. 3, in accordance with various embodiments, a cross-section of a piston assembly **300** is illustrated. Piston assembly **300** includes similar components to those described above with respect to piston assembly **200** in FIG. 2, including a piston **302**, a spring guide **304**, a spring **306**, a spring sleeve **308**, a first insulator **310**, an insulator shield **312**, a lock ring **316**, an insulator support **318**, and a component **320**. Piston assembly **300** further includes a second insulator **322** that is located between component **320** and insulator shield **312**. Various features and components of piston assembly **300** are described above with respect to piston assembly **200** and FIG. 2 which may not be repeated here.

(30) Second insulator **322** is made from a high temperature capable insulator material. In various embodiments, second insulator **322** may be made from carbon. In various embodiments, second insulator **322** may be made from austenitic nickel-chromium-based superalloys such as those sold under the name INCONEL(IN-100) or another high temperature capable material. In the depicted embodiment, first insulator **310** has a different cross-section than insulator **210** depicted in FIG. 2. The cross-section of first insulator **310** includes a shelf portion **310a** upon which second insulator **322** is disposed. Additionally, insulator shield **312** has an annular shape including a central hole, or opening, through which second insulator **322** extends to contact heat shield **314**. Insulator shield **312** contacts first insulator **310** but does not contact second insulator **322**. In various embodiments, a portion of insulator shield **312** may contact first insulator **310**.

(31) There is a distance $d2$ between second insulator **322** and second load path part **320**. Distance $d2$ provides a thermal barrier between second insulator **322** and second load path part **320** to reduce heat transfer between the two components. In various embodiments, distance $d2$ may be about 0.1 mm to about 1 mm. In various embodiments distance $d2$ may be larger or smaller based on piston assembly **300** design. Similar to piston assembly **200** described above with respect to FIG. 2, in the event of high temperatures, such as during a HBRT event, second insulator **322** and/or first insulator **310** may collapse. Second load path part **320** provides additional support for piston assembly **300** in response to the collapse with little to no additional heat being introduced to piston assembly through second load path part **320**.

(32) Referring now to FIG. 4, in accordance with various embodiments, a cross-section of a piston assembly **400** is illustrated. Piston assembly **400** includes similar components to those described above with respect to piston assembly **200** in FIG. 2, including a piston **402**, a spring guide **404**, a spring **406**, a spring sleeve **408**, an insulator **410**, an insulator shield **412**, a lock ring **416**, an insulator support **418**, and a first component **420**. Piston assembly **400** further includes a second component **424**. Various features and components of piston assembly **300** are described above with respect to piston assembly **200** and FIG. 2 which may not be repeated here.

(33) Second component **424** may be an annular shaped part embedded in piston **402**, as illustrated. In various embodiments, second component **424** fits within a channel, or groove, in piston **402**. Second component **424** serves as an additional secondary load path to first component **420** by providing additional support for spring guide **406** in response to piston assembly **400** collapsing due to extreme temperatures. As described above, when piston assembly **400** collapses, distance $d1$ disappears and insulator shield **412** directly contacts first component **420** to maintain braking pressure. The forces on first component **420** are transferred to spring guide **404** which may cause springs **406** to compress thereby reducing braking pressure applied by piston assembly **400**. Second component **424** provide additional support to spring guide **404** and spring **406** to prevent compression of spring **406** in a HBRT event. In various embodiments, spring guide **404** may contact second component **424** prior to first component **420** being engaged. In various, embodiments, there may be an air gap between second component **424** and spring guide **404**. The air gap acts as a thermal barrier between second component **424** and spring guide **404**, reducing heat transfer. Additionally, the air gap provides space for spring guide **404** to move (e.g., in the y-axis) in response to normal braking condition without engaging second component **424**.

(34) Referring now to FIG. 5, in accordance with various embodiments, a cross-section of a piston assembly **500** is illustrated. Piston assembly **500** includes similar components to those described above with respect to piston assembly **300** in FIG. 3 and piston assembly **400** in FIG. 4, including a piston **502**, a spring guide **504**, a spring **506**, a spring sleeve **508**, a first insulator **510**, an insulator shield **512**, a lock ring **516**, an insulator support **518**, a first component **520**, a second insulator **522**, and a second component **524**. Various features and components of piston assembly **500** are described above with respect to piston assembly **300** and FIG. 3 and piston assembly **400** in FIG. 4 and may be repeated here.

(35) In the depicted embodiment, piston assembly **500** includes first insulator **510** and second insulator **522** as described above with respect to piston assembly **300** in FIG. 3. Piston assembly **500** further includes first component **520** and second component **524** as described above with respect to piston assembly **400** in FIG. 4. The combination of these components allows piston assembly **500** to maintain brake pressure for a longer period of time during a HBRT0 event than would be otherwise possible. This improves the safety of piston assembly **500** and the brake assembly (e.g., brake assembly **100**).

(36) Benefits, other advantages, and solutions to problems have been described herein with regard to specific embodiments. Furthermore, the connecting lines shown in the various figures contained herein are intended to represent exemplary functional relationships and/or physical couplings between the various elements. It should be noted that many alternative or additional functional relationships or physical connections may be present in a practical system. However, the benefits, advantages, solutions to problems, and any elements that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as critical, required, or essential features or elements of the disclosure. The scope of the disclosure is accordingly to be limited by nothing other than the appended claims, in which reference to an element in the singular is not intended to mean “one and only one” unless explicitly so stated, but rather “one or more.” Moreover, where a phrase similar to “at least one of A, B, or C” is used in the claims, it is intended that the phrase be interpreted to mean that A alone may be present in an embodiment, B alone may be present in an embodiment, C alone may be present in an embodiment, or that any combination of the elements A, B and C may be present in a single embodiment; for example, A and B, A and C, B and C, or A and B and C. Different cross-hatching is used throughout the figures to denote different parts but not necessarily to denote the same or different materials.

(37) Systems, methods and apparatus are provided herein. In the detailed description herein, references to “one embodiment,” “an embodiment,” “various embodiments,” etc., indicate that the embodiment described may include a particular feature, structure, or characteristic, but every embodiment may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases are not necessarily referring to the same embodiment. Further, when a particular feature, structure, or characteristic is described in connection with an embodiment, it is submitted that it is within the knowledge of one skilled in the art to affect such feature, structure, or characteristic in connection with other embodiments whether or not explicitly described. After reading the description, it will be apparent to one skilled in the relevant art(s) how to implement the disclosure in alternative embodiments.

(38) Numbers, percentages, or other values stated herein are intended to include that value, and also other values that are about or approximately equal to the stated value, as would be appreciated by one of ordinary skill in the art encompassed by various embodiments of the present disclosure. A stated value should therefore be interpreted broadly enough to encompass values that are at least close enough to the stated value to perform a desired function or achieve a desired result. The stated values include at least the variation to be expected in a suitable industrial process, and may include values that are within 10%, within 5%, within 1%, within 0.1%, or within 0.01% of a stated value. Additionally, the terms “substantially,” “about” or “approximately” as used herein represent an amount close to the stated amount that still performs a desired function or achieves a desired

result. For example, the term “substantially,” “about” or “approximately” may refer to an amount that is within 10% of, within 5% of, within 1% of, within 0.1% of, and within 0.01% of a stated amount or value.

(39) Furthermore, no element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(40) Finally, it should be understood that any of the above described concepts can be used alone or in combination with any or all of the other above described concepts. Although various embodiments have been disclosed and described, one of ordinary skill in this art would recognize that certain modifications would come within the scope of this disclosure. Accordingly, the description is not intended to be exhaustive or to limit the principles described or illustrated herein to any precise form. Many modifications and variations are possible in light of the above teaching.

Claims

1. A piston assembly, comprising: a piston having a first end, an opposing second end, and a sidewall extending from the first end to the second end, the piston further including a cavity at least partially defined by the sidewall; a spring guide having a first end and an opposing second end, the spring guide disposed within the cavity of the piston and adjacent the sidewall; an insulator having a first portion extending from the second end of the piston and a second portion disposed within the piston; an insulator shield disposed adjacent the first portion of the insulator; and a component coupled to, surrounding, and enclosing the second end of the spring guide between the spring guide and the insulator shield, the component being offset from the insulator shield by a distance thereby forming an airgap between the component and the insulator shield.
2. The piston assembly of claim 1, wherein the insulator is a first insulator, the piston assembly further comprising: a second insulator disposed adjacent the second end of the first insulator, wherein the insulator shield includes an opening through which the second insulator extends.
3. The piston assembly of claim 1, wherein the component is a first component, the piston assembly further comprising: a second component disposed between the first end of the spring guide and the first end of the piston and coupled to the sidewall of the piston, the second component configured to provide support to the spring guide.
4. The piston assembly of claim 3, wherein the second component comprises an insulator material.
5. The piston assembly of claim 1, wherein the component comprises an insulator material.
6. The piston assembly of claim 1, wherein the component is press fit onto the second end of the spring guide.
7. The piston assembly of claim 1, further comprising: a spring disposed between the first end of the spring guide and the first end of the piston; a lock ring disposed in the sidewall of the piston and configured to prevent the spring guide from extending out of the second end of the piston; and an insulator support disposed within the piston and configured to secure the insulator to the piston.
8. A brake assembly, comprising: a plurality of stator disks; a plurality of rotor disks interleaved with the plurality of stator disks; a pressure plate disposed adjacent one of the plurality of rotor disks; and a piston assembly disposed adjacent the pressure plate, the piston assembly including: a piston having a first end, an opposing second end, and a sidewall extending from the first end to the second end, the sidewall at least partially defining a cavity; a spring guide having a first end and an opposing second end, the spring guide disposed within the cavity of the piston and adjacent the

sidewall; an insulator having a first portion extending from the second end of the piston and a second portion disposed within the piston; an insulator shield disposed adjacent the first portion of the insulator adjacent the pressure plate; and a component coupled to, surrounding, and enclosing the second end of the spring guide between the spring guide and the insulator shield, the component being spaced from the insulator shield by a distance thereby forming an airgap between the component and the insulator shield.

9. The brake assembly of claim 8, wherein the insulator is a first insulator and the piston assembly further comprises: a second insulator disposed adjacent the second end of the first insulator, wherein the insulator shield includes an opening through which the second insulator extends.

10. The brake assembly of claim 9, wherein the second insulator is configured to engage the pressure plate to apply a braking force.

11. The brake assembly of claim 8, wherein the component is a first component and the piston assembly further comprises: a second component disposed between the first end of the spring guide and the first end of the piston and coupled to the sidewall of the piston, the second component configured to provide support to the spring guide.

12. The brake assembly of claim 11, wherein the second component comprises an insulator material.

13. The brake assembly of claim 8, wherein the component comprises an insulator material.

14. The brake assembly of claim 8, wherein the component is threaded onto the second end of the spring guide.

15. The brake assembly of claim 8, wherein the component is press fit onto the second end of the spring guide.

16. The brake assembly of claim 8, wherein the piston assembly further comprises: a spring disposed between the first end of the spring guide and the first end of the piston; a lock ring disposed in the sidewall of the piston and configured to prevent the spring guide from extending out of the second end of the piston; and an insulator support disposed within the piston and configured to secure the insulator to the piston.

17. The brake assembly of claim 8, wherein the insulator shield is configured to engage the pressure plate to apply a braking force.

18. The brake assembly of claim 8, wherein the component is configured to engage the pressure plate in response to the insulator shield collapsing.

19. A piston assembly, comprising: a hollow cylindrical piston having a first end and an opposing second end; a spring guide having a first end and an opposing second end disposed within the hollow cylindrical piston; a component coupled to, surrounding, and enclosing the second end of the spring guide; an insulator coupled to the second end of the hollow cylindrical piston, the insulator offset from the component by a first distance; and an insulator shield coupled to the insulator and offset from the component by a second distance thereby forming an airgap between the component and the insulator shield, wherein the component is disposed between the spring guide and the insulator shield.

20. The piston assembly of claim 19, further comprising: a second insulator coupled to the insulator and offset from the component by a third distance; and a second component coupled to the piston and configured to provide support for the spring guide.
